MIL-C-60945A(AR)
15 September 1980
SUPERSEDING
MIL-C-60945(MU)
11 January 1968

# MILITARY SPECIFICATION
# CARTRIDGE, 40MM, PRACTICE, M385; PROJECTILE METAL PARTS FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers the metal parts for one type of cartridge designated as Cartridge, 40MM, Practice, M385.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extend specified herein:

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-A-8625 | - | Anodic Coatings for Aluminum and Aluminum Alloys |
| MIL-A-48078 | - | Ammunition, Standard Quality Assurance Provisions, General Specification For |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-414 | - | Sampling Procedures and Table for Inspection by Variables For Percent Defective |

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**Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.**

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS (See 6.5)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8866714 - Projectile Assembly, For Cartridge, 40MM, Practice, M385

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117 73 - Standard Method of Salt Spray (Fog) Testing

(Applications for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing (Dwg) 8866714, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a through, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stampling shall be neat and sharply defined.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection. (See 3.3)

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

| Part Description | Drawings | Quantity |
|---|---|---|
| Body | 8866715 | 25 |
| Projectile Assembly | 8866714 | 25 |

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.



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TABLE I. FIRST ARTICLE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

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| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Projectile Assembly and Components | | | | SEE BELOW |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Body (Dwg. 8866715) | | | | |
| | Examination for defects | 25 | - | | 4.4.2.1 |
| | Salt Spray | 5 | - | | 4.5.1 |
| | Projectile Assembly (Dwg. 8866714) | | | | |
| | Examination for defects | 25 | | | 4.4.2.2 |
| | Push test of rotating band | 10 | | | 4.5.2 |

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete



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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables sampling plans for major and minor defects shall be in accordance with MIL-STD-105.



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1 | Body | | 8866715 |
| | | | NEXT HIGHER ASSEMBLY 8866714 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL: | None defined | | | | |
| MAJOR | | | | | |
| 101. | Diameter of rotating band surface (after knurl) | | 0.40% | 3.2 | Gage |
| 102. | Location of basic diameter on spherical radius | | 0.40% | 3.2 | Gage |
| 103. | Concentricity of indicated area on spherical radius with body diameter | | 0.40% | 3.2 | Gage |
| 104. | Location of basic diameter | | 0.40% | 3.2 | Gage |
| 105. | Concentricity of indicated area on taper with body diameter | | 0.40% | 3.2 | Gage |
| 106. | Body diameter | | 0.40% | 3.2 | Gage |
| 107. | Diameter between knurl | | 0.40% | 3.2 | Gage |
| 108. | Width of crest of knurl | | 0.40% | 3.2 | Gage |
| 109. | Width of rotating band surface | | 0.40% | 3.2 | Gage |
| 110. | Length from back of rotating band surface to front end of node | | 0.40% | 3.2 | Gage |
| 111. | Salt Spray | * | | 3.2 | 4.5.1 |
| MINOR | | | | | |
| 201. | Protective coating with bare spot on bottom or forward of rotating band groove | | 0.65% | 3.2 | Visual |
| 202. | Incorrect color | | 0.65% | 3.2 | Visual |
| 203. | Evidence of poor workmanship | | 0.65% | 3.4 | Visual |

NOTES:

* Sampling and rejection shall be in accordance with MIL-A-8625.



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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

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| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.2 | Projectile Assembly | | | | 8866714 |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL: | None defined | | | | |
| MAJOR | | | | | |
| 101. | Location of bourrelet shoulder | | 0.40% | 3.2 | Gage |
| 102. | Diameter of bourrelet | | 0.40% | 3.2 | Gage |
| 103. | Profile of rotating bands | | 0.40% | 3.2 | Gage |
| 104. | Diameter of rotating bands | | 0.40% | 3.2 | Gage |
| 105. | Diameter of large groove | | 0.40% | 3.2 | Gage |
| 106. | Diameter of small groove | | 0.40% | 3.2 | Gage |
| 107. | Diamter below bourrelet | | 0.40% | 3.2 | Gage |
| 108. | Distance from nose to back of rotating band | | 0.40% | 3.2 | Gage |
| 109. | Concentricity of rotating bands with bourrelet | | 0.40% | 3.2 | Gage |
| 110. | Total length | | 0.40% | 3.2 | Gage |
| MINOR | | | | | |
| 201. | Evidence of poor workmanship | | 0.65% | 3.4 | Visual |
| 203. | Marking missing | | 0.65% | 3.2 | Visual |

NOTES:



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4.4.3 Testing.

4.4.3.1 Push test of rotating band on projectile assembly. Beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have been accepted, a sample of fifty (50) assemblies in accordance with MIL-STD-414, Table B-I, with an AQL of 0.25 shall be selected from each band setting machine, each lot, and subjected to this test.

NOTE: Movement as specified on Dwg. 8866714 is defined as movement of the back face of the rotating band relative to the body.

4.4.3.2 Push test of rotating band on projectile assembly. After three (3) acceptable lots have met the criteria of 4.3.3.1, the Government inspectior shall select twenty-five (25) assemblies in accordance with MIL-STD-414, Table B-I with an AQL of 0.25 from each band setting machine, each lot and subjected to the test. If any lot fails the above criteria the lots thereafter shall revert to the criteria of 4.3.3.1.

NOTE: Movement as specified on Dwg. 8866714 is defined as movement of the back face of the rotating band relative to the body.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Salt spray. This test shall be conducted in accordance with ASTM B 117-73 for 24 hours using 5% salt solution. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.2 Push test of rotating band on projectile assembly. The projectile assembly shall be placed in an approved fixture and the force specifieid on the drawing shall be applied. Observation shall be made for the requirement specified on the drawing. This test is a destructive test. Parts so tested shall not be returned to the lot.

NOTE: Deformation of the chamfered leading edge of the rotating band caused by compression in the test fixture is permissible.



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5. PACKAGING

5.1 Preservation and packing.

5.1.1 Level C. Preservation and packaging shall be in accordance with MIL-STD-1169.

5.2 Packing.

5.2.1 Level C. The projectile assembly preserved and packaged as specified in 5.1.1, shall be packed in accordance with MIL-STD-1169.

5.3 Marking. Marking shall comply with MIL-STD-1169.

5.4 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent missing of the lots in transit.

6. NOTES

6.1 Intended use. The components covered by this specification is intended for use on Cartridge, 40MM, Practice, M385.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, NJ 07801. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Submission of results of contractor - conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARRADCOM, ATTN: DRDAR-QAA-S, Dover, NJ 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.



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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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